

RESEARCH ARTICLE

Impact of Generative Modeling for Fundus Image Augmentation With Improved and Degraded Quality in the Classification of Glaucoma

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ABSTRACT Glaucoma is a heterogeneous group of diseases characterised by cupping of the optic nerve head and visual field damage, starting with a progressive loss of vision that leads to permanent blindness. When diagnosed in time, Glaucoma can be delayed by adequate treatment. More efficient processes for diagnosis are being proposed, and the role of artificial intelligence in the field is growing. This work presents a pipeline to evaluate the impact of generative modelling in Computer-Aided Diagnosis (CADx) of Glaucoma based on Deep Learning, particularly focused on the optic disc region. The methodology relies on transforming retinal fundus images to improve and degrade their quality to augment the training data and assess the diagnostic performance. The objective evaluation of the proposed model based on Generative Adversarial Networks revealed quantitative and qualitative improvements in image quality. To support this, we propose a new model to evaluate the quality of fundus images, which can also be used within the pipeline to reject samples with lower image quality for diagnosis. Its performance surpassed related work, achieving a balanced accuracy of 0.929. Concerning Glaucoma CADx, the results obtained in public datasets point to a considerable gain in Sensitivity, Specificity, and Accuracy, achieving scores of 0.883 (+0.054), 0.957 (+0.019), and 0.931 (+0.031), respectively, after image data augmentation when compared with previous work targeted at offline inference in mobile devices. Considering the restriction of choosing simpler backbone networks that can run on edge devices, our findings support the importance of image quality diversity and realistic augmentation.

INDEX TERMS Computer-aided diagnosis, convolutional neural networks, deep learning, fundus images, generative adversarial networks, glaucoma, image quality, image classification, retina.

I. INTRODUCTION

Glaucoma is the most frequent cause of irreversible blindness worldwide. Its prevalence is projected to rise with population growth and ageing [1], with an estimated increase of people affected by Glaucoma from 76 million people (40 to 80 years old) in 2020 to 112 million by 2040 [2]. However, the disease

is still underdiagnosed as it is usually detected by opportunistic case finding [3].

Despite being irreversible, Glaucoma can be effectively slowed down with appropriate treatment, pharmaceutical or surgical, reducing the risk of complications [4]. Therefore, early diagnosis is of utmost importance, resulting in a demand for new diagnostic techniques to aid in the case analysis process, rendering Glaucoma assessment more efficient.

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Glaucoma can be considered a heterogeneous group of diseases characterised by cupping of the optic nerve head and visual field damage. Thus, structural analysis by retinal imaging is crucial to cover most of its types. Chronic forms of Glaucoma are painless, and symptomatic visual field defects occur late [5]. Fundus images can be used to evaluate morphological features to diagnose several eye conditions, including Glaucoma.

This work contributes to a more decentralized Computer-Aided Diagnosis of Glaucoma by proposing a pipeline to evaluate the impact of generative-based augmentation for training the CADx system, able to run offline on edge devices. The methodology relies on transforming retinal fundus images to improve and degrade their quality, augment the data, and assess the classification performance.

The CADx system tackles a binary classification of Glaucoma (normal or suspicious glaucomatous optic nerve) based on a small field-of-view centred in the optic disc. Our mobile-targeted solution showed promising results, improving upon previous work, as evaluated on an aggregate of relevant public datasets. Moreover, we propose a new model to evaluate the quality of fundus images outperforming the related work, which can be used within the pipeline to reject samples with lower image quality for diagnosis.

A. MACHINE LEARNING IN GLAUCOMA CADx

The candidates for Glaucoma annotation structures include but are not limited to the following [6]: optic disc - a central round-like yellowish part and the entry point for vessels; optic cup - located inside the optic disc, it is a bright central depression with variable size; peripapillary atrophy (PPA) - appears as a crescent-shaped part of the eye, composed of an alpha-zone and a beta-zone outside the optic disc border, being the beta-zone closest to the disc. Such zones' size, shape, and texture are important for analysis [7]. Therefore, Glaucoma annotation in retinal images is mainly based on the fundus morphology of these structures. The clinical observation range is usually a local circular region with a diameter around two optic discs on the fundus colour photography. Accordingly, image segmentation and classification tasks should include a wider region around the optic disc [8], [9].

More recently, Deep Learning approaches have seen a tremendous increase in research as they tend to bypass the need for hand-crafted features. Deep Residual Learning for Image Recognition (ResNet) [10] have been one of the most popular backbones for binary classification of glaucoma [11], [12], [13], [14]. In [11], [13], [15], and [16], several pre-trained CNN architectures were fine-tuned to the Glaucoma classification problem, concluding that models from other domains show competitive performance when fine-tuned.

A recent work by Mahrooqi et al. [17] propose a multiview classification. Different EfficientNet-B0-based networks are trained independently in different views of the retinal fundus: the original image resized; the cropped region around the optic disc; and the polar transformation of the cropped

region. The AUC of 0.92 is achieved by combining the top models of these training sessions, giving more weight to the best-performing one which is not consistent among datasets.

Another approach presented by Nawaz et al. [18] proposes the EfficientDet-D0 as a detection model to identify the optic disc and optic cup glaucomatous regions for classification, reporting an average accuracy of 0.98 but in a limited number of datasets publicly available while ignoring relevant regions around the optic nerve head [7], [8], [9].

Martins et al. [19] created a classification network with MobileNetV2 as a feature extractor backbone, followed by a global average pooling layer and two fully connected layers, interleaved by heavy dropouts. This architecture obtained results similar to state-of-the-art approaches, albeit with fewer parameters, targeting inference on edge devices to promote decentralized CADx of Glaucoma.

Contrary to the data typically used to train these models, which is well standardized and captured with high-end equipment, real-world data bring limitations such as poor and variable image quality. Moreover, demanding computational requirements may hinder the deployment, especially when complex architectures are used. Data-centric approaches should be explored in these scenarios, namely generative-based augmentation or image quality enhancement.

B. QUALITY EVALUATION IN RETINA

Low-quality retina images can complicate the clinical analysis or result in unreadable cases, hindering the performance of CADx tools or human reading and labelling. One of the first research works focused on the quality evaluation of retinal images for automated grading of Diabetic Retinopathy, was proposed by Fleming et al. [20]. The authors consider two factors: image clarity (e.g., affected by poor camera focus or bad optics, saccadic eye motion during acquisition, cataracts, macular edema); the field definition (e.g., caused by poor patient orientation, insufficient pupil size, or latent squint) as the image obtained should show the correct area of the retina with visible optic disc and temporal arcades [20]. In Niu et al. [21], these factors were expanded to account for the presence of artefacts besides the visibility of the optic disc and macula. In Costa et al. [22], the image quality was evaluated using Q_v score by computing a local degree of vesselness around each pixel, and Image Structure Clustering metrics focused on a global evaluation, and not solely on vessels, which shows the worst behaviour on synthetic images.

Several studies have developed neural networks capable of automatically assessing the quality of retina images and thus evaluating which are suitable for medical diagnosis. The EyeQual model developed by Costa et al. [23] was trained on a Multiple Instance Learning framework limited to binary labels and validated in small Diabetic Retinopathy datasets. The Multiple Color-space Fusion Network (MCF-Net) proposed by Fu et al. [24] relied on more than 28792 retinal images in the EyeQ dataset and three quality levels.

The methods in [20] and [21] are not enough to evaluate the retinal image quality since they are highly dependent on homogeneous light conditions during acquisition. On the other hand, the work of Fu et al. [24] shows better performance while employing dense architectures, which makes its implementation more challenging, namely when deployed in edge devices such as handheld fundus cameras.

C. SYNTHETIC DATA GENERATION FOR RETINAL IMAGES

Retinal fundus images are expensive to obtain and involve patient privacy issues, resulting in a complex research subject. Many researchers have applied Generative Adversarial Networks (GANs) to expand the image datasets [25], alleviating the problem of insufficient data and uneven sample distribution. Creating synthetic data can be more efficient and cost-effective than collecting real-world data in many cases. It can also be created on-demand based on specifications rather than waiting to collect it once it occurs in reality. Synthetic data can complement real-world data so that testing can happen for every imaginable variable, even if there are no good examples in the original data set.

Guibas et al. [26] proposed a pipeline to generate fundus images for a segmentation task end-to-end using a pair of GANs. The first GAN focuses on varying the geometry of the segmentation mask, and the second GAN transforms the geometry into a photorealistic image.

Costa et al. [27] presented a method where instead of learning a transformation between the vessel trees and the corresponding retinal image, it pairs the original vessel map and their corresponding retinal fundus images to train an autoencoder model. Then, it could learn how to generate new data from a given vessel map. Nevertheless, it depends on the vessel extraction method (e.g., a general U-net architecture [28]), where the quality of the segmented vessel tree ultimately affects the generated retinal image.

Shen et al. [29] proposed a discriminative and enhancement model to correct low-quality fundus images, proposing a degradation model to generate high-/low-quality image pairs for training. The proposed network improves the disc/cup ratio estimation and vessel segmentation performance. A relatively close work by Son et al. [30] focused on image segmentation, introducing a GAN to produce more accurate and precise vessel maps compared with the state-of-art methods, showing outstanding results on fine vessels.

Some problems with training this type of network are mode collapse and unstable training. Output control is key, especially in complex datasets where the best way to ensure accurate outputs is by comparing synthetic data with authentic or human-annotated data. This is due to possible inconsistencies in synthetic data when trying to replicate complexities within original datasets. A more general approach would be the class conditional-aware adversarial models as in [31].

D. SYNTHETIC DATA GENERATION FOR GLAUCOMA

Although the literature presents an excellent performance in Glaucoma classification, there is a significant issue of

generalization to new cases. The number of images labelled for Glaucoma available in public datasets is scarce. As such, image synthesis methods can be found in the state-of-art related to Glaucoma detection.

Bonaldi et al. [32] used a model that produces the retinal background and the fovea, and another model to generate the optic disc relying on a large dictionary of patches with no vessels that are later registered and blended. In Diaz-Pinto et al. [33] in 2018, VAE and DCGAN architectures were trained on cropped retinal fundus images from six different datasets. The authors evaluate Cohen's kappa coefficient from ten experts with 3 to 10 years of experience on 100 synthetic images and 100 real images. The results show that most experts were deceived when evaluating if the image was authentic or generated. The exception was the two most experienced evaluators. The authors compared the real and artificial images through statistical analysis of relevant structures, such as the vessels and optic disc, concluding that the pixel proportions were similar. Both the approaches in [32] and [33] generate completely new images, requiring further labelling of the cases even though the retinal images may look realistic. Alternative approaches are needed to perform controlled data augmentation while preserving the significant risk factors detectable in the structural analysis of the retina. To the best of our knowledge, no studies use generative modelling to slightly manipulate existing fundus images in the area of Glaucoma.

The use of a GAN on Glaucoma progression using Optical Coherence Tomography (OCT) scans was studied by Hassan et al. [34] in 2020. The authors explore image sequences to predict the outcome on a fourth visit, using the first two or three visits. The results suggest that only two prior visits may be sufficient to predict the following OCT scan. The total number of images used is 26592 cross-sections of 101 glaucomatous eyes [34]. Although some artefacts are generated, the work presents interesting methods. However, it is applied to OCT scans, whose current availability is not comparable with the widespread, cheaper fundus cameras that are becoming more portable.

E. IMAGE DATA AUGMENTATION

AutoAugment [35] is an automated approach to find data augmentation policies from data. It formulates the problem of finding the best augmentation policy as a discrete search problem since one of the challenges of constructing refined labels for post-augmented data is considering the augmentation's safety. It relies on a Reinforcement Learning algorithm that searches for an optimal policy amongst a constrained set of geometric transformations with various distortion levels. However, it is somewhat domain-dependent, proving a challenge for developing generalizable augmentation policies. RandAugment [36] is a stochastic data augmentation routine for image data, composed of strong augmentation transforms like colour jitter, Gaussian blur, and saturation, along with more traditional augmentation transforms such as random crops. It also achieves higher accuracy on a wide range of

benchmarks and scales well in ImageNet or COCO datasets, thanks to its ability to adjust its distortion magnitude according to the model and dataset size. However, more studies are needed to tailor the transformations to a given task, expand the routine to include bounding box-specific transformations or evaluate the benefits in other domains where data augmentation improves predictive performance.

F. EVALUATING ARTIFICIALLY GENERATED IMAGES

The last years have witnessed the rise of a set of objective metrics for GAN evaluation [37], such as the Inception Score (IS), the Frechet Inception Distance (FID) [38], or the recent Likeness Score (LS) [39]. Three aspects are expected in a good GAN-based model [39]: Creativity (non-duplication of real images); Inheritance (style fidelity of generated images, retaining the key features of real ones); and Diversity (generated images should be different from each other).

Whereas the first metrics (IS and FID) depend on a pre-trained Inception network, LS evaluates the generated images directly. Hence, the former two metrics present notable limitations in considerably different domains where retinal images can be included. For instance, the IS does not consider real data (and thus, the Creativity aspect). This means that if the generated images deviated from the original data but got closer to the images from the dataset used to train the Inception network (e.g., ImageNet), the IS would be misleadingly higher. The FID depends on a pre-trained Inception network based on the difference between authentic and generated image distributions. It assumes a multivariate Gaussian distribution of the feature vectors, which is often not assured. Conversely, LS is based on the separability between the two classes (real vs generated images), which is computed considering the intra-class and between-class distance sets [39]. This allows to evaluate the three discussed aspects, making it a better candidate to evaluate GANs objectively. However, in domain transfer applications, the set of real vs generated samples must be carefully chosen to consider both domains, which is not always easy. Thus, expert visual inspection or another type of qualitative analysis is still preferential in several applications [37].

II. METHODOLOGY

The methodology of this research can be divided into three main modules. Section II-C presents the proposed GAN-based synthetic retinal image generation method for data augmentation, which aims at enhancing the performance of Glaucoma CADx systems. In section II-B, the suggested retinal image quality evaluator for assessing the quality of the generated images is introduced. Finally, we study the impact of using the data augmentation methods presented in II-C, and also its combination with the retina quality evaluator to reject low-quality images on the training of a Computer-Aided Diagnosis model for Glaucoma. Figure 1 presents a block diagram summarizing the proposed methodology.

TABLE 1. Quantitative description of the EyeQ dataset regarding image quality levels.

Quality level	Training Samples	Test Samples	Total samples
Good	8347	8470	16817
Usable	1896	4559	6435
Reject	2320	3220	5540
Total	12543	16249	28792

TABLE 2. Collections of datasets used to evaluate the image quality manipulation models (DS1). Columns indicate the name of the dataset, the number of positives (glaucoma), the number of negatives (normal), and the type of image. There are two type of images.

Layer	Positives	Negatives	Total	Type of image
ORIGA [40]	168	482	650	Full FOV
DrishTi-GS [41]	70	31	101	Full FOV
REFUGE [14]	80	720	800	Full FOV
Total	318	1233	1551	-

A. DATASETS AND PRE-PROCESSING

To the best of our knowledge, there are no publicly available datasets with both annotations of image quality and the presence of Glaucoma. Therefore, two distinct datasets were compiled to address both aspects separately.

For the development and training of the image quality assessment and transformation models, we used the EyeQ dataset [24], a quality-related annotation of a subset of the EYEPACS dataset summarized in Table 1, including labels regarding image quality. The quality standard was based on the diagnosability for general eye diseases, such as Glaucoma, Age-related Macular Degeneration or others, rather than only Diabetic Retinopathy. These labels differentiate image quality according to a three-level quality grading system:

- ‘Good’ - the retinal image presents no low-quality factors and all pathology-related features are clearly visible;
- ‘Usable’ - some low-quality indicators are visible yet the lesions are clear enough to be identified by ophthalmologists;
- ‘Reject’ - the retinal image has serious quality issues impeding a full and reliable diagnosis.

As the ‘Reject’-labelled images present insufficient quality for diagnosis, we decided to consider only the intermediate quality level images (‘Usable’) for building the low-quality domain set. According to the authors, two experts in eye fundus labeled each image from the EYEPACS dataset individually, creating two labels for each image. Those where these labels were not equal were discarded, ensuring the reliability of the remaining labels.

A second dataset (DS1) was then compiled for assessing the performance of our image quality manipulation models when using data from sources different than the EyeQ dataset. This was achieved by combining datasets with images associated with Glaucoma labels and presenting the

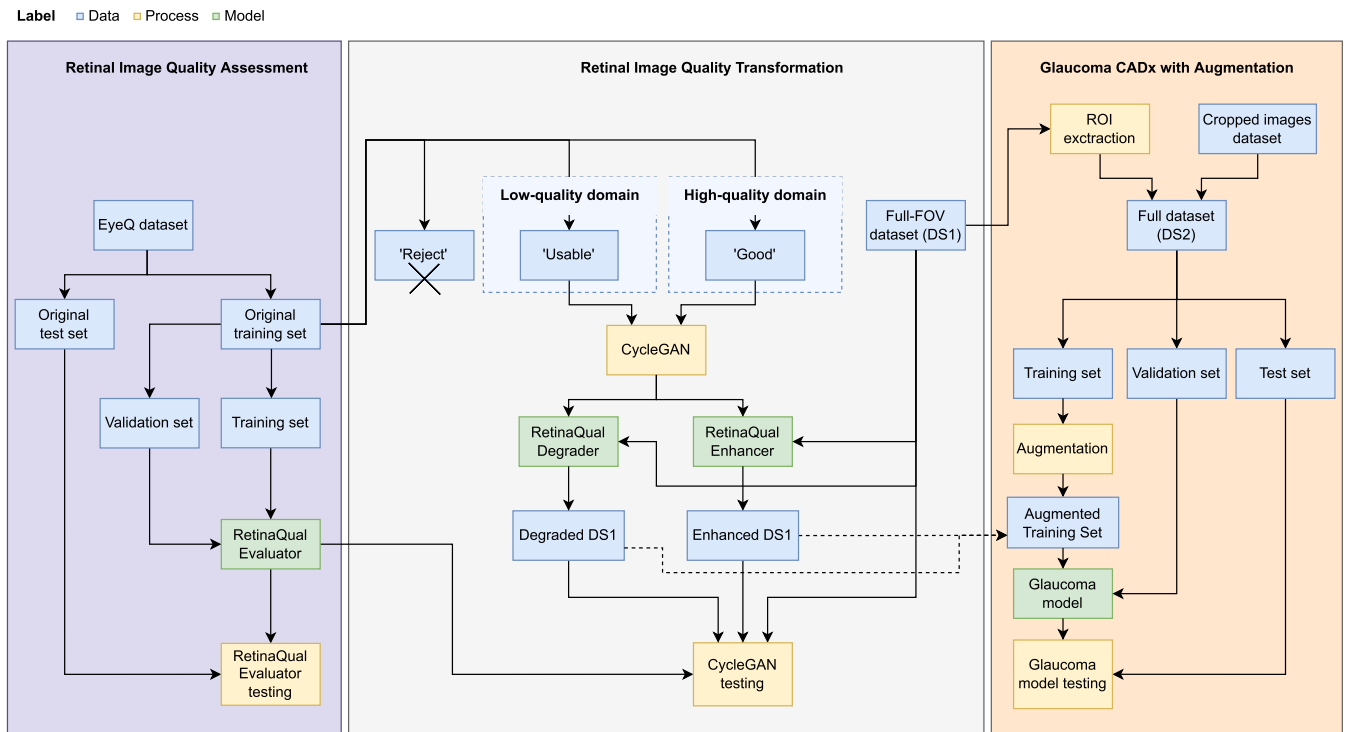


FIGURE 1. Block diagram of the proposed methodology.

TABLE 3. Collections of datasets used to train and evaluate the Computer-Aided Diagnosis models for Glaucoma (DS2), including those that present the full FOV of the fundus image and those that only contain cropped regions of the optic disc (OD).

Layer	Positives	Negatives	Total	Type of image
ORIGA [40]	168	482	650	Full FOV
Drishti-GS [41]	70	31	01	Full FOV
REFUGE [14]	80	720	800	Full FOV
RIM-ONE r1 [42]	40	118	158	OD crops
RIM-ONE r2 [43]	255	200	455	OD crops
RIM-ONE r3 [43]	148	170	318	OD crops
ACRIMA [16]	396	309	705	OD crops
Total	1157	2030	3187	-

full field-of-view (FOV) image taken with the fundus camera instead of a cropped region around the optic disc. The dataset is presented in Table 2.

Finally, a third dataset (DS2) was formed to evaluate the effects of image quality manipulation on the performance of a CADx model for Glaucoma. This dataset was obtained by compiling multiple publicly available datasets containing Glaucoma labels, presented in Table 3. Since some datasets include images showing the full FOV and others contain only images cropped around the area of the optic disc, the samples in DS2 were harmonized by cropping as applied in [19].

B. RETINA QUALITY EVALUATOR

The Retina Quality Evaluator (RetinaQualEvaluator) algorithm aims to distinguish between the three image quality

TABLE 4. Architecture of the convolutional neural network used for training RetinaQualEvaluator.

Layer	Units	Activation	Output Shape
EfficientNetB0	-	-	[7, 7, 1792]
Global Average Pooling	-	-	[1792]
Fully Connected	16	Leaky ReLU	[3]
Fully Connected	1	None	[1]

classes mentioned ('Good', 'Usable' and 'Reject'). It takes as input an uncropped fundus image with a full FOV such as those provided by the EyeQ dataset. To train our RetinaQual-Evaluator, and because EyeQ does not have, originally, a validation set, we extracted 500 images from the original training set (around 4%) to use as a validation set, using the remaining samples for training the model.

The algorithm starts by cropping the black margins and padding the image to inscribe it in a square image. This image is then resized to a resolution of 224×224 as in section II-C. This image is then used as input for a Convolutional Neural Network that employs the convolutional part of an EfficientNetB0 [44] pre-trained on ImageNet using Noisy Student Training [45] as a backbone model. The generated feature maps are then followed by Global Average Pooling and two Fully Connected layers. The structure of the network can be found in Table 4. To reduce overfitting, L1 and L2 regularization of 0.01 each and Dropout of 0.3 were applied to the first Fully Connected layer. The model's output is a single

value that follows the labels provided by the EyeQ dataset, i.e., 0 for ‘Good’, 1 for ‘Usable’, and 2 for ‘Reject’. Note that the model can output values such as 0.5 or 1.7, as the quality scale can be considered continuous.

The model was trained using the ADAM optimization algorithm with a learning rate of 0.0001, a β_1 of 0.9, and a β_2 of 0.999. The loss function was mean squared error, suitable for a regression task. To account for class imbalance, class weights were set using the method proposed by King and Zheng [46]. For augmentation, only random horizontal and vertical flips are performed to minimize the impact on the quality of the images in the training data.

C. SYNTHETICALLY AUGMENTED DATA GENERATION

To generate the synthetic retinal images for augmentation, we propose using a Cycle Generative Adversarial Network (CycleGAN) architecture. This Deep Learning framework allows Unpaired Image-to-Image translation, which consists of an image synthesis task involving the generation of a new image that is a supervised modification of an input image. Because the training process is done without paired examples, the goal is to learn a mapping $G: X \rightarrow Y$ such that the distribution of images from $G(X)$ is as identical to the distribution Y as possible. This is accomplished by using an adversarial loss [47].

The transformation process requires a source domain X and a target domain Y . The image domains considered in this project are low- and high-quality. The low-quality set images are characterized by displaying defects such as blurring, uneven illumination, over/under-exposure, or flash-induced artefacts, which are commonly observed in real-world acquisition settings. These conditions are particularly detectable when using portable and inexpensive imaging devices. On the other hand, the high-quality domain images present few or none low-quality factors, and all morphological structures, as well as pathology-related features, are clearly visible. As such, we define two domains:

- **HI:** High-quality domain, represented by the images labeled as ‘Good’ in the EyeQ dataset;
- **LO:** Low-quality domain, represented by the images labeled as ‘Usable’ in the EyeQ;

As the ‘Reject’-labelled images present insufficient quality for diagnosis, we decided to consider only the intermediate quality level images (‘Usable’) for building the low-quality domain set. Using these domains to synthesize retinal images for augmentation allows a CADx training set with more realistic quality heterogeneity.

The CycleGAN architecture is composed of two generators and one discriminator. The generators carry out the translation process into the opposite domain while the discriminator evaluates the authenticity of the generated images, that is, how similar the translated images are to the target domain. The framework utilizes two CycleGANs with reversed directions which are trained as illustrated in Figure 2. While the generators are shared by the two architectures, the

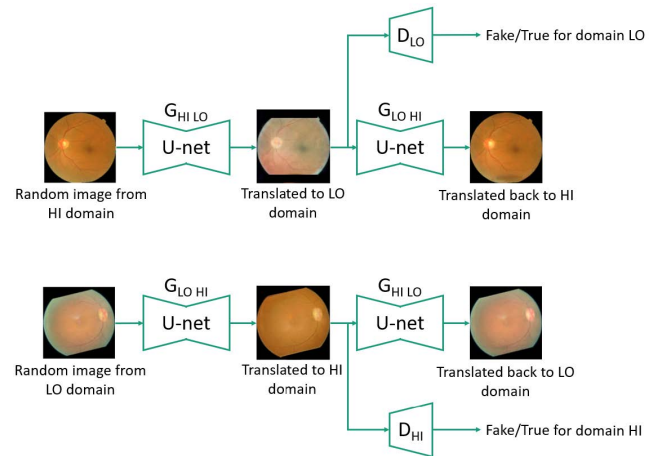


FIGURE 2. CycleGAN training illustration in which domain HI and LO correspond to high- and low-quality respectively. For each CycleGAN, the process begins with a set of random images of one domain, these images are translated to the target domain; then they are evaluated for their authenticity; and finally translated back to the original domain.

discriminators are distinct as they evaluate different domain authenticities. Hence, the system is composed of four neural networks in total. Although the inner architectures for the Generators and Discriminators can be adapted, and their parameters changed, the standard training of CycleGANs is followed.

Images from both domains are first pre-processed by cropping the black margins and padding the image to inscribe it in a square image. This is then resized to a resolution of 224×224 , obtaining a smaller image with a preserved aspect ratio, and normalized from -1 to 1 . These images are used to train the generators $G_{LO \rightarrow HI}$ and $G_{HI \rightarrow LO}$, which translate images from low- to high-quality domain and from high- to low-quality domain, respectively. Each generator is formed by a U-Net, which consists of a convolutional neural network that contains a contraction section with successive convolutions and an expansive section with successive deconvolutions. These are performed using Downsampling and Upsampling blocks, respectively. These are defined as:

- **Downsampling Block:** Composed of a 2D convolutional layer with a filter size of 4 and a stride of 2, a LeakyReLU activation and instance normalization.
- **Upsampling Block:** Composed of an upsampling layer, a convolutional layer with a filter size of 4 and a stride of 2, instance normalization and finally a concatenation in the channel axis with the output of the Downsampling Block with identical height and width.

In the expansive path, each upsized image is concatenated with the corresponding image from the contracting path. This allows the network to combine both the positional and the contextual information. In Table 5 we present in detail the architecture of the employed generators. The discriminators (D_{HI} and D_{LO}) are PatchGAN discriminator networks that evaluate the validity of all patches composing an image, outputting a 2D array with local authenticities [48]. Thus, this method considers different sections of the image for the

TABLE 5. Architecture of generators of the CycleGAN.

Layer	Units	Activation	Output Shape
Downsampling Block	64	LeakyReLU	[112, 112, 64]
Downsampling Block	128	LeakyReLU	[56, 56, 64]
Downsampling Block	256	LeakyReLU	[28, 28, 64]
Downsampling Block	512	LeakyReLU	[14, 14, 64]
Upsampling Block	256	LeakyReLU	[28, 28, 64]
Upsampling Block	128	LeakyReLU	[56, 56, 64]
Upsampling Block	64	LeakyReLU	[112, 112, 64]
Convolutional 2D	3	tanh	[224, 224, 3]

TABLE 6. Architecture of discriminators of the CycleGAN. Each discriminator block consists of a 2D convolution with a filter size of 4 and a stride of 2, a LeakyReLU activation and instance normalization.

Layer	Units	Activation	Output Shape
Discriminator Block	32	LeakyReLU	[112, 112, 64]
Discriminator Block	64	LeakyReLU	[56, 56, 64]
Discriminator Block	128	LeakyReLU	[28, 28, 64]
Discriminator Block	256	LeakyReLU	[14, 14, 64]
Convolutional 2D	Sigmoid		[14, 14, 1]

assessment. The network comprises multiple discriminator blocks, consisting of a 2D convolution with a filter size of 4 and a stride of 2, a LeakyReLU activation, and instance normalization.

In Table 6 we present in detail the architecture of the employed discriminators. To minimize overfitting, a dropout of 0.2 is introduced after each hidden layer during training in both generators and discriminators. The discriminators are trained separately from the generators as they are used to assess the latter's performance during the learning process. The loss function for the discriminators corresponds to a mean square error loss. For the CycleGAN framework, the loss function is given by

$$Loss_{total} = \sum_{n=\{A,B\}} (w_r Loss_{r_n} + w_i Loss_{i_n} + w_{id} Loss_{id_n})$$

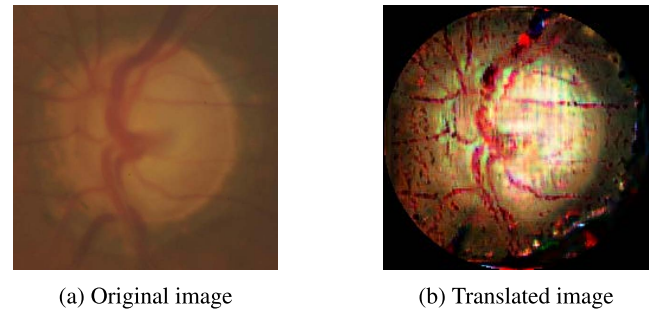
$Loss_{r_n}$ corresponds to mean absolute error functions for the reconstruction of the translated images. This assures cycle-consistency, which implies that the underlying information of the image is preserved while only the domain of the image changes. $Loss_{i_n}$ are mean square error loss functions for the validation of the translated images through the discriminator. The validation loss allows the generators to produce images that could realistically have been drawn from the target domain set. Lastly, the $Loss_{id_n}$ functions correspond to mean absolute error functions forcing the images to equal their original form after passing through the generator to assure that the identity of the images is maintained. Intuitively, the latter's weight loss needs to be considerably inferior to the first two to guarantee a successful domain translation. The weights are set to $w_r = 10$, $w_i = 1$ and $w_{id} = 0.1$.

For data augmentation in the training of both the discriminators and CycleGANs, we applied vertical and horizontal flipping, rotation in a range of 0-20 degrees, and a slight

brightness variation in a range of 0.9-1.1. Upon training, the generators G_{LOHI} and G_{HILO} are isolated and applied in generating synthetic image sets in both considered domains. For easier reference, the models for improving and degrading the quality of fundus images will be referred to as RetinaQualEnhancer and RetinaQualDegradator, respectively.

D. CADx IN GLAUCOMA

Most of the CADx algorithms designed for Glaucoma, such as those presented in our prior work [19], are trained using cropped regions from around the optic disc. Since the previous synthetic generative models are trained on the EyeQ dataset, they are designed to function with uncropped images with a wide field-of-view, such as those presented in Figure 5. Therefore, when applying our augmentation models, their behaviour is undefined and thus unpredictable. Figure 3 shows an example of such transformation.

**FIGURE 3.** Result of applying the GAN-based quality improvement model to a cropping of the optic disc without any preprocessing. As the image does not present the typical black borders existing in the training set, the model presents an unpredictable behaviour.

To allow using these models to cropped regions of the optic disc, we apply the following steps:

- 1) Pad the image by replicating the border to obtain an image of 336×336 .
- 2) Multiply the image by a circular mask, obtaining an image resembling a full fundus such as in Figure 4b.
- 3) Apply the model to the resulting image, obtaining an image similar to Figure 4c.
- 4) Crop the image back to the original region, obtaining an image similar to Figure 4d.

We start by reproducing the work of Martins et al. [19], where a classification network was built using the Original data split in 80/10/10 train, test and validation (see Table 7) with stratification and random seed fixed without any image transformation and based on a MobilenetV2 backbone. We compare these results to the ones obtained using a different backbone model, the EfficientNetB0 [44], also trained and tested with the original version of dataset DS2, the one used in this set of experiments. Then, classic image augmentation methods were applied. These consist of flipping the image and randomly applying one of three types of blur (Gaussian, median, and average). Additionally, we changed the image contrast, brightness, and sharpness, with a 0.5 probability

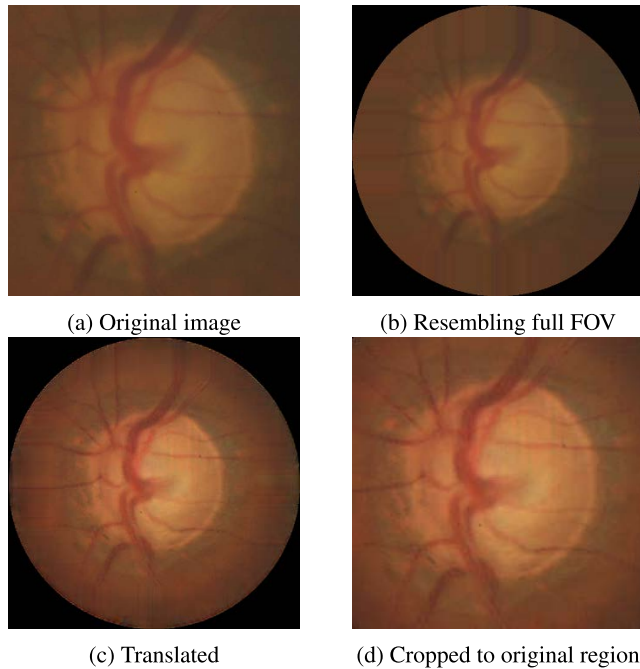


FIGURE 4. Result of applying the GAN-based quality improvement model to a cropping of the optic disc, after the described preprocessing pipeline. (a) shows the original image, (b) the image manipulated to resemble a fundus image without cropping to the region of the optic disc, (c) the result of applying RetinaQualEnhancer and (d) the image cropped to restore the original bounds of the image.

of picking any such operation, to gauge its impact on the performance. Moreover, a set of experiments was performed, taking only EfficientNetB0 as the backbone model. These include variations on the training data or the testing set. All the networks for the classification were trained through a singularity container in a high performance computing cluster equipped with a 48-core Intel® Xeon® Silver 4214 CPU (2.20GHz) and a 32 GB NVIDIA® V100 GPU after resizing the images to 224×224 and performing data normalization. The multi-label dice loss was selected, together with the ADAM optimizer with a learning rate of 0.0001. Additionally, the learning rate was reduced during the learning process by a factor of 2 if the validation loss was not improved in the last 10 epochs. Furthermore, the model was trained for 200 epochs, with early stopping if no improvements are verified in the validation loss after 20 epochs to mitigate overtraining.

III. RESULTS AND DISCUSSION

A. EVALUATING THE QUALITY OF RETINAL IMAGES

As aforementioned, the capability of evaluating the quality of retinal images has different important applications. A reliable quality evaluator can be used, for instance, in the acquisition step to ensure higher-quality images are collected, possibly asking the clinician to repeat the acquisition when a sample is classified as having lower-quality. Following the objective of this work, the authors considered the hypotheses that the impact of using synthetically generated retinal images for

TABLE 7. DS2 dataset partition and stratification used to train and evaluate the Computer-Aided Diagnosis models for Glaucoma.

DS2 partitions	Positives	Negatives	Total
Train	860	1689	2549
Validation	131	188	319
Test	111	208	319
Total	1102	2085	3187

data augmentation may be higher, if the augmented samples could be filtered to keep only higher-quality images. Moreover, when transforming retinal images to enhance their quality, one can also apply the quality evaluator to reject samples that have their quality downgraded instead. However, such applications require a quality evaluator yielding high confidence results.

Concerning the new proposed RetinaQualEvaluator, it addresses some of the identified issues with the former architecture. The standard classification approach disregards that the class labels represent ordered levels, and thus the model does not weigh differently a ‘Good’ sample classified as ‘Usable’ or ‘Reject’, which is not optimal. As such, treating the problem as a regression task and including the Quadratic Weighted Kappa as an evaluation metric helps to mitigate this problem.

Table 8 shows the results of the proposed RetinaQualEvaluator using the original test set, compared with the ones obtained for the MCF-Net [24], both the published results and those obtained by using the provided code. Our run of the MCF-Net allowed us to get scores for the different classes instead of the aggregated ones presented by Fu et al. [24]. Only slight differences were observed, possibly due to small changes in the implementation and data samples.

When comparing these results with the proposed RetinaQualEvaluator, our approach presents better metrics overall, except Precision against Xu et al. method [51], and slightly on Recall for Muddamsetty et al. [50]. Comparing with the MCF-Net, the proposed network is correct more often, and the errors it makes are less severe (opposed to classifying a ‘Reject’ image as ‘Good’). Overall, we are confident that our network with an EfficientNetB0 backbone for evaluating the quality of eye fundus images improves or is at least competitive with the state-of-the-art.

B. VISUALIZING QUALITY TRANSFORMATIONS

As described in Section I-F, especially for specific domains such as retinal image analysis, visual analysis is still preferred when evaluating the quality of images resulting from generative models. The ultimate evaluation goal is to have a panel of expert ophthalmologists visually inspect our results and grade them according to their quality to assess if an improvement was actually taking place when applying our model.

Nevertheless, in Figure 5 we present a set of visual examples from dataset DS1, to support a qualitative analysis and demonstrate the effects of the proposed model in image

TABLE 8. Summary of the results for retinal image quality assessment, comparing our RetinaQualEvaluator to other available methods using EyeQ's original test set. Since the code for the method proposed by Fu et al. [24] is publicly available, we decided to execute it to extract additional metrics.

Method	Accuracy	Balanced Accuracy	Quadratic Kappa	Precision	Recall	F1-Score
Fu et al. [24]	-	0.918	-	0.865	0.850	0.855
Fu et al. (our run)	0.880	0.920	0.896	0.865	0.857	0.861
Raj et al. [49]	0.884	-	-	0.869	0.870	0.869
Muddamsetty et al. [50]	-	-	-	0.878	0.880	0.878
Xu et al. [51]	0.890	-	-	0.890	0.872	0.872
Zhou et al. [52]	-	0.920	-	0.866	0.870	0.868
RetinaQualEvaluator (Ours)	0.894	0.929	0.912	0.879	0.878	0.878

quality translation, both for improving and degrading the retinal image quality. Looking at the samples, we can see differences in image quality when comparing the second column (Improved) with the first (Original) and third (Degraded) ones. Improvements can be observed regarding vessels and optic disc definition and/or contrast (across all rows) and the correction of some illumination artefacts (third and sixth rows especially). The Degraded versions (third column) show less pronounced transformations when compared with the original images. Still, we can see the inclusion of some artefacts (darker spot in the fourth row) or less contrast on the optic disc and vessels (fifth row).

We will refrain from discussing the impact on specific Glaucoma-related diagnostic features, as they would depend on validation by a panel of expert ophthalmologists. However, this subjective visual analysis still shows the potential of such models in improving image quality. To further evaluate this potential, we proceed with more objective measures, be it at the quality level and its impact on CADx for Glaucoma, presented in the following subsections.

C. EVALUATING THE QUALITY OF TRANSFORMED IMAGES

Due to the limitations presented in Section I-F, we do not use usual metrics of evaluating GANs, as IS and FID, since the application domain is different from the classes used to pre-train the Inception model underlying such methods. From a different perspective, if we had a perfect retinal image quality evaluator available, we could use it to assess the quality of image samples before and after transformation (quality improvement or degradation). Unfortunately, such an evaluator does not exist, although we can rely on the proposed RetinaQualEvaluator to generalize how the transformation affects image quality.

Figure 6 shows the probability distribution of the RetinaQualEvaluator predictions regarding a continuous quality level (where 0 is Good, 1 is Usable, and 2 is Reject) for the original, improved and degraded versions of the retinal images under study, comprising dataset DS1. As expected, one can observe a clear shift when comparing the transformed versions to the original one. The probability of predictions closer to 0 (Good) increases for the improved version

(orange) and decreases for the degraded version (green). On the other hand, the probability of predictions closer to 2 (Reject) drops for the improved transformation, whereas an increase is observed for the degraded version. This result is especially interesting since Reject samples were not used to train the transformation models, which could still transform enough features of the image to change a considerable number of images from the Reject class closer to Usable and Good. Even though the RetinaQualEvaluator is not a perfect evaluation tool for this purpose, it showed enough performance and generalization ability so that this result objectively supports the proposed model to transform the quality of retinal images (in both directions - improvement and degradation).

D. EVALUATING THE IMPACT OF TRANSFORMED IMAGES ON CADx FOR GLAUCOMA

Ultimately, the goal of developing an image quality enhancer surpasses the visual assessment and quality evaluation by itself. The real value of such a tool lies in the possible improvements it might bring to diagnostic models. CADx systems based on ML rely heavily on the amount and quality of training data. These should be representative of the domain at hand (the pathology to diagnose), with sufficient examples and variability to generalize as well as possible to new, unseen data given as input to the system when in production.

In this context, we assess the impact of our image transformation models on building a CADx model for Glaucoma risk assessment, as summarized in Table 9. One of the goals was to validate whether the proposed tool could be used as a standalone filter in screening. To this end, we considered using our image transformation model as an enhancer filter, where all training and testing data were put through the transformer model to improve image quality. Then, the model was used for data augmentation in training rather than as a filter. In this approach, we considered three different scenarios, adding different transformed sets to the original data: a) only Improved-quality images; b) Improved- and Degraded-quality images; and c) both Improved- and Degraded-quality Images, removing the transformed images that are classified as Rejected by the trained RetinaQualEvaluator.

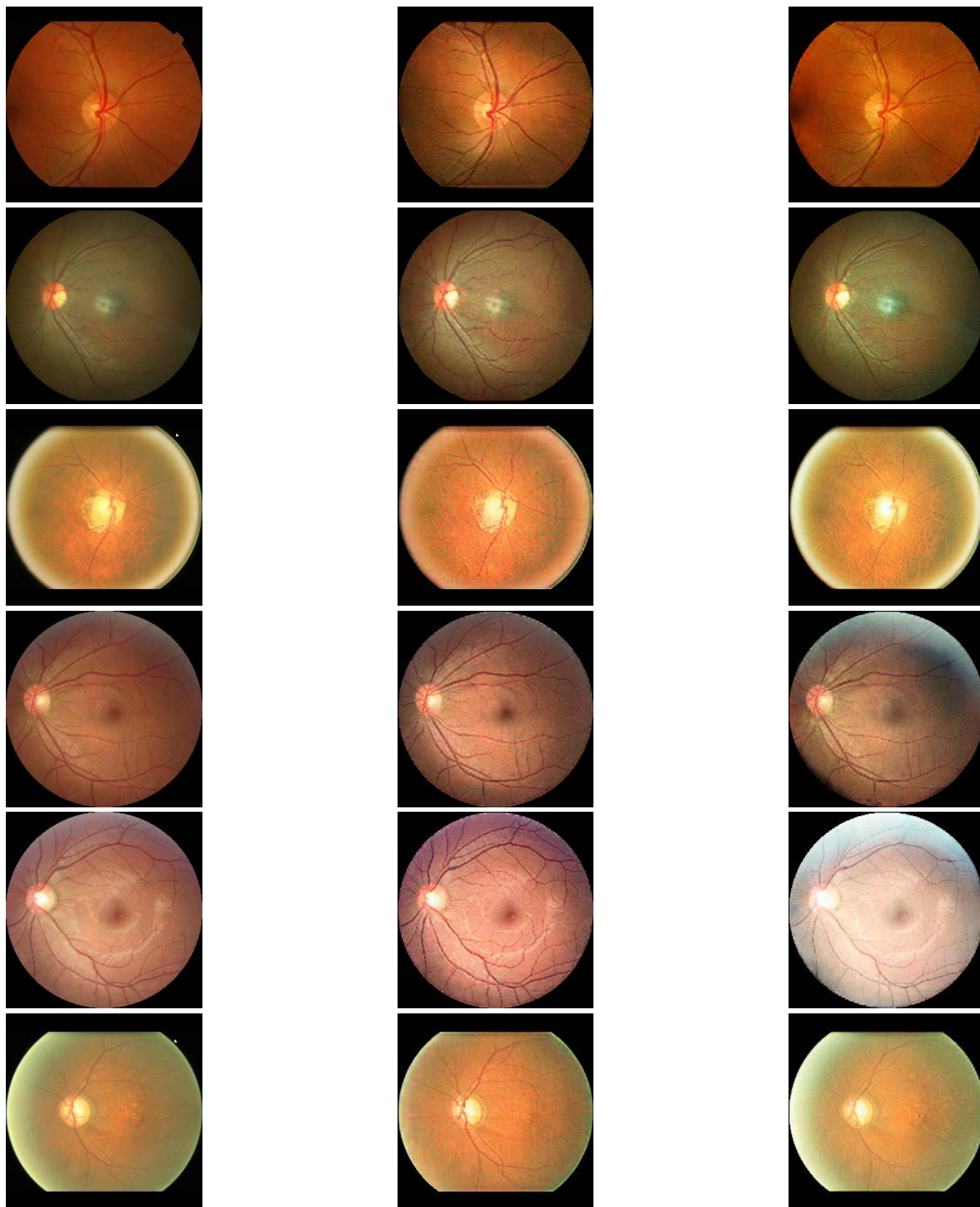


FIGURE 5. Sample results from the application of the proposed Cycle-GAN model for transforming retinal image quality using dataset DS1, from which a qualitative analysis can be derived. Left - Original image, Center - Improved quality, Right - Degraded quality.

TABLE 9. Classification performance of Glaucoma CADx results obtained on the test set using fundus images, including the impact of applying the proposed image quality transformer. *Orig* is the original dataset of retinal images as described in Section II-A; *Classic Augment.* represented synthetically generated images through geometric augmentation techniques; *Improved* and *Degraded* are the transformed version of the retinal images using the transformer for increasing and decreasing image quality, respectively.

Base Model	Training	Testing	Sensitivity	Specificity	Accuracy
MobilenetV2	Orig	Orig	0.829	0.938	0.900
EfficientNetB0	Orig	Orig	0.766	0.981	0.906
MobilenetV2	Orig + Classic Augment.	Orig	0.847	0.894	0.878
EfficientNetB0	Orig + Classic Augment.	Orig	0.838	0.942	0.915
EfficientNetB0	Improved	Improved	0.864	0.928	0.906
EfficientNetB0	Orig + Improved	Orig	0.883	0.942	0.922
EfficientNetB0	Orig + Improved + Degraded	Orig	0.883	0.957	0.931
EfficientNetB0	Orig + Improved + Degraded - (Rejected)	Orig	0.866	0.931	0.910

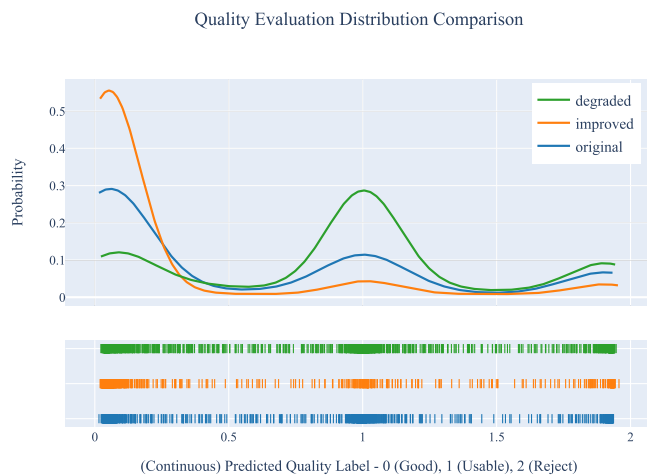


FIGURE 6. Probability distribution of quality levels (as classified by our RetinaQualEvaluator) for the original, improved and degraded versions of the images, obtained through the use of the proposed GAN-based RetinaQualEnhancer (and RetinaQualDegrader) on images from dataset DS1.

The first rows of Table 9 show that classic image augmentation, contrary to what was expected, did not improve the overall results for a MobilenetV2 backbone model (a decrease can be seen in both Specificity and Accuracy scores). Meanwhile, an improvement is observed for the EfficientNetB0 base model (an increase of 7% in Sensitivity 1% in Accuracy scores). Comparing both backbone models, we can see that EfficientNetB0 presents better results, especially when classic augmentation is used. These results supported the decision to proceed with the EfficientNetB0 backbone model for the remainder of the analysis.

Regarding the use of our transformation model as a filter versus using it as a Data Augmentation technique, we can see that, across all considered metrics, it's more impactful when considering it for augmenting the training set. Even compared to the original training (second row in Table 9), although we gain in Sensitivity, Specificity is decreased, resulting in a very similar Accuracy score. This finding suggests that, even though we could observe an improvement of the image quality either visually and at the quality evaluation level,

the transformation might make some disease-specific characteristics more salient while hindering others. On the other hand, enriching the dataset with the transformed versions seems the best choice for this CADx system. Within the Data Augmentation scenarios, we can observe that more data appears to be associated with a better performance. In fact, adding both Improved- and Degraded-quality versions of the images to the original data yields the best results across all metrics, reaching a Sensitivity over 95% and Accuracy a little over 93%. Somewhat unexpectedly, we can observe that removing the set of images classified by the proposed Retina Quality Evaluator as Rejected from the previous set (Original + Improved + Degraded) actually worsens the results. The last results and the visual inspection of the Degraded images (see Figure 5) can support these findings. The transformation to lower-quality images is often not that significant, adding minor artefacts that will not hinder the diagnostic performance but still increase the variability of the training data, indeed helping the model generalize to new images. As the Retina Quality Evaluator only considers a general quality level, not Glaucoma-related features, we are, in fact, decreasing the input data and the network's generalization performance by removing 'Reject' samples.

IV. CONCLUSION

The quest for decentralized healthcare, addressing challenges concerning early screening such as the lack of diagnostic equipment or trained professionals, is drawing more and more attention in the last years. Nonetheless, making equipment more portable and low-cost usually results in lower data quality, namely regarding retinal image acquisition. Thus, bridging this gap by improving image quality would definitely support this goal by providing specialists with better quality images and enriching and enhancing CADx systems built upon these data.

This work proposes a new framework for retinal image quality improvement, as acquired by eye fundus cameras, able to transform lower-quality samples into better-quality ones using Deep Learning generative models. A crucial step of this pipeline concerns evaluating whether the image quality

was actually improved, kept, or degraded. Moreover, such a method could be used as a stand-alone filter to reject lower-quality images, even at the acquisition stage, prompting the user to get a new, higher-quality image. To this end, and to include an objective evaluation metric besides visual inspection, we propose a new model called RetinaQual-Evaluator based on a lightweight EfficientNetB0 network to classify retinal images according to their quality level (a continuous level from 0 - 'Good', 1 - 'Usable' up to 2 - 'Reject'). We note that this evaluator can be applied to different eye-related pathologies using retinal fundus images. The regression-based approach and the used backbone network allowed us to surpass the state-of-the-art results achieving scores of 0.894 for Accuracy, 0.912 for Quadratic Weighted Kappa, 0.879 for Precision, 0.878 for Recall, and 0.878 for F1-Score.

The ultimate purpose of the proposed framework for image quality transformation is to improve CADx systems, in this case, for the diagnosis of Glaucoma. Hence, we evaluated the impact of the proposed transformation network in several ways: using it as a filter, at both training and testing stages, using only the Improved version of the images; or using it as a data augmentation strategy, adding the Improved (and Degraded) samples to the training dataset. The latter was shown to be the best option, achieving metrics of 0.883 for Sensitivity, 0.957 for Specificity, and 0.931 Accuracy, using an EfficientNetB0 model backbone. Considering the restriction of choosing simpler backbone networks that can run on edge devices, such as smartphones, our findings support the importance of image quality diversity and realistic augmentation, contributing to the effective decentralized Computer-Aided Diagnosis of Glaucoma, improving upon previous work [19].

Concerning future work, we recall that both the generative model and the quality evaluator were trained with images of full field-of-view, while different equipment and datasets present retinal images with lower field-of-view. We wish to address this by generalizing the training steps of both models to different field-of-views. Furthermore, we will present the results of enhancing (and possibly degrading) image samples to a panel of expert ophthalmologists to validate our image quality improvement framework and expand its application to other domains. These include other ophthalmological diseases, but even other medical imaging applications, as we believe these generative approaches, used to filter lower-quality acquisitions or enrich AI-based diagnostic models, can have a tremendous positive impact on decentralized healthcare.

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